

ACM Summer School on Symmetric Key Cryptography

Introduction to Cryptography: Part 2

Department of Computer Science and Engineering

Indian Institute of Technology Hyderabad

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Outline

- 1 Information-Theoretic Security
- 2 Computational Security
- 3 Security Against Eavesdroppers



What is Cryptography?

Cryptography is the science of securing information and communication.

Main goals:

- 1 Confidentiality
- 2 Integrity
- 3 Authentication
- 4 Non-repudiation

Applications:

- HTTPS
- Banking systems
- Cloud computing
- Messaging systems
- Blockchains
- Secure operating systems



From Classical to Modern Cryptography

Classical cryptography relied largely on:

- Hiding the algorithm
- Statistical confusion
- Human difficulty in analysis

Modern cryptography asks:

Can security be defined mathematically?

Two major approaches:

- 1 Information-theoretic security
- 2 Computational security



Information-Theoretic Security

An encryption scheme is information-theoretically secure if security holds even against an adversary with

- Unlimited computation
- Unlimited memory
- Unlimited time

Security depends solely on the information available to the adversary.



Probability Review

Let

$$M$$

denote the random variable representing the message and

$$C$$

the random variable representing the ciphertext.

Cryptographic security is often expressed using conditional probabilities such as

$$\Pr[M = m \mid C = c].$$



Definition (Shannon)

An encryption scheme achieves perfect secrecy if

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

for every message m and ciphertext c .

Interpretation:

Observing the ciphertext does not change the adversary's belief about the message.



Equivalent Definition

Perfect secrecy is equivalent to

$$\Pr[C = c \mid M = m_0] = \Pr[C = c \mid M = m_1]$$

for all

$$m_0, m_1 \in \mathcal{M}$$

and every ciphertext

$$c \in \mathcal{C}.$$

(related to previous definition via Bayes' theorem)

Thus every message induces exactly the same ciphertext distribution.



One-Time Pad

Invented by Gilbert Vernam.

Key properties:

- Chosen uniformly at random
- Key length equals message length
- Key used exactly once

Encryption:

$$c = m \oplus K$$

Decryption:

$$m = c \oplus K.$$



One-Time Pad Example

Message:

$$m = 10110110$$

Key:

$$K = 01100101$$

Ciphertext:

$$c = m \oplus K$$

$$c = 11010011.$$



Why OTP Works

Given ciphertext c and message m , there exists exactly one key

$$K = c \oplus m.$$

Thus for every message m ,

$$\Pr[C = c | M = m] = \Pr[K = c \oplus m].$$

Since K is uniformly distributed,

$$\Pr[C = c | M = m] = 2^{-n}$$

for every message.

Hence the ciphertext distribution is independent of the message.



Limitations of Perfect Secrecy

The One-Time Pad is theoretically optimal but:

- Long keys are required
- Keys must be truly random
- Keys cannot be reused
- Key distribution is difficult

Motivation for computational security.



What Happens if the Key is Reused?

Suppose the same key encrypts two messages:

$$C_1 = M_1 \oplus K$$

$$C_2 = M_2 \oplus K.$$

An attacker intercepts both ciphertexts and computes

$$C_1 \oplus C_2.$$

Substituting:

$$(M_1 \oplus K) \oplus (M_2 \oplus K).$$

Using

$$K \oplus K = 0,$$



Why This Breaks Perfect Secrecy

The attacker does not immediately learn M_1 or M_2 ,
but learns

$$M_1 \oplus M_2.$$

Suppose the attacker correctly guesses part of M_1 .
Then

$$(M_1 \oplus M_2) \oplus M_1 = M_2.$$

A correct guess about one plaintext reveals the corresponding part of the other plaintext.

Therefore the ciphertexts leak information about the messages:

$$P(M = m \mid C = c) \neq P(M = m).$$

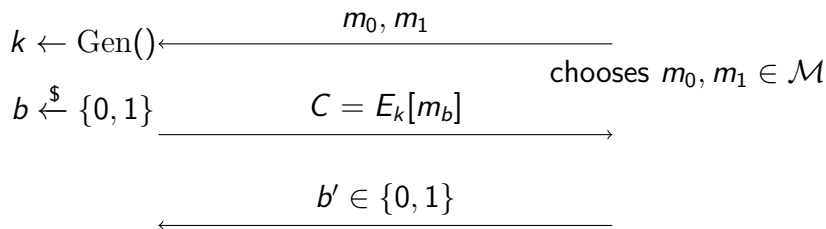
Perfect secrecy is lost.



Perfect Secrecy as a Game PrivK

Challenger

Adversary



$$\text{PrivK}^{\text{eav}} = \begin{cases} 1 & \text{if } b = b', \\ 0 & \text{if } b \neq b'. \end{cases}$$

$$\Pr[\text{PrivK}^{\text{eav}} = 1] = \frac{1}{2} \quad \text{for every adversary.}$$



Why is perfect secrecy is impractical?

Modern systems protect:

- Gigabytes of data
- Billions of users
- Global communication networks

Perfect secrecy is often impractical.

Instead we assume adversaries have limited computational resources.



Two relaxations:

- Security preserved only against computationally bounded (efficient) adversaries
- Adversaries succeed with very small probability.
- A scheme is (t, ϵ) -secure if every adversary running for time at most t succeeds in breaking the scheme with probability at most ϵ .
- Eg $(t, t/2^n)$ security: If key length is 256 and $t = 2^{100}$, then $\epsilon = \frac{1}{2^{156}}$



Security Parameter

- Modern cryptography is parameterized by λ called the security parameter.
- Intuitively, the key length K depends on λ .
- The cryptosystem itself runs in $\text{poly}(n^\lambda)$ (size of the security parameter).
- As λ increases:
 - Security increases
 - Computational cost increases



Probabilistic Polynomial-Time (PPT)

A PPT algorithm:

- Uses randomness
- Runs in polynomial time
- Represents realistic attackers

Throughout modern cryptography:

Adversaries are modeled as PPT algorithms.



Negligible Functions

A function

$$\mu : \mathbb{N} \rightarrow \mathbb{R}$$

is negligible if

$$\forall c > 0, \exists N$$

such that

$$n > N \Rightarrow \mu(n) < \frac{1}{n^c}.$$



Examples of Negligible Functions

Negligible:

$$2^{-n}$$

$$2^{-\sqrt{n}}$$

$$\frac{1}{n^{100}} \quad (\text{for sufficiently large fixed exponent})$$

Not negligible:

$$\frac{1}{n}$$

$$\frac{1}{\log n}.$$



Symmetric Key Cryptosystem

Now, three algorithms:

- ① $Gen(1^\lambda) = K$
- ② $Enc_k(m) = c$
- ③ $Dec_k(c) = m$
- ④ such that $\forall m, k, Dec_k(Enc_k(m)) = m$



Computational Indistinguishability

Intuition:

Two distributions look identical to every efficient observer.

Central concept in modern cryptography.



Formal Definition

Random variable ensembles

$$X_\lambda$$

and

$$Y_\lambda$$

are computationally indistinguishable if

$$|Pr[D(X_\lambda) = 1] - Pr[D(Y_\lambda) = 1]|$$

is negligible for every PPT distinguisher D .



Distinguishing Advantage

Define

$$\mathit{Adv}_D(\lambda) = |\mathit{Pr}[D(X_\lambda) = 1] - \mathit{Pr}[D(Y_\lambda) = 1]|.$$

Security requires

$$\mathit{Adv}_D(\lambda) = \mathit{negl}(\lambda).$$

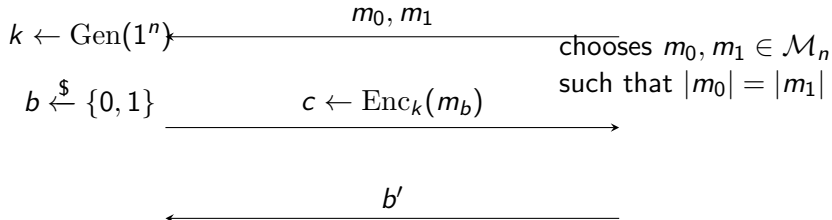


Computational Security: $\text{PrivK}^{\text{eav}}$ Experiment

A PPT adversary $\mathcal{A}$ and an encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$

Challenger

Adversary



$$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = \begin{cases} 1 & \text{if } b = b', \\ 0 & \text{otherwise.} \end{cases}$$



Π is EAV-secure if for every PPT adversary $\mathcal{A}$ there exists a negligible function $\text{negl}(\cdot)$ such that

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

Notice the mention of PPT adversary and $\text{negl}(n)$ that wasn't there in the perfect secrecy.



Two Views of Cryptographic Security

Information-Theoretic Security

- Security against an all-powerful adversary.
- Measured using probability and entropy.
- Introduced by Claude Shannon.

Computational Security

- Security against efficient adversaries.
- Adversaries are modeled as PPT algorithms.
- Measured using distinguishing advantage.

Shannon: Information

Modern Crypto: Computation



In the language of Shannon Entropy

Let M be a random variable representing the message.

- The (self) Information of the outcome $M = m$ is $I(M) = -\log_2 P(M = m)$
- and the Shannon entropy of M is $H(M) = -\sum_m P(M = m) \log_2 P(M = m)$.

Interpretation of Entropy:

- Measures uncertainty in the message.
- Larger entropy means more uncertainty.
- Entropy is measured in bits.

After observing a ciphertext C , the remaining uncertainty is

$$H(M | C).$$



Perfect Secrecy and Entropy

An encryption scheme has perfect secrecy if

$$P(M = m \mid C = c) = P(M = m)$$

for all messages m and ciphertexts c .

This implies

$$H(M \mid C) = H(M)$$

Observing the ciphertext does not reduce uncertainty.

Equivalently, in the language of *mutual information* (measures how much information one random variable contains about another)

$$I(M; C) = H(M) - H(M \mid C) = 0.$$



The Limitation of Entropy-Based Security

Entropy measures how much information exists.

Modern cryptography asks a different question:

Can an efficient adversary exploit that information?

Even if some information is present: $I(M; C) > 0$, it may still be impossible for a polynomial-time algorithm to use it effectively.

Therefore modern cryptography focuses on *computational distinguishability* rather than entropy alone.



Entropy vs. PPT Security

Shannon Security	Computational Security
All-powerful adversary	PPT adversary
Entropy-based	Game-based
$H(M C) = H(M)$	Advantage is negligible
$I(M; C) = 0$	IND-CPA, IND-CCA, etc.
Perfect secrecy	Practical security

Information-Theoretic $\implies$ Computational

but not vice versa.



Modern cryptography studies progressively stronger adversaries:

- ① Eavesdropper (IND-EAV)
- ② Known Plaintext Attack (KPA)
- ③ Chosen Plaintext Attack (CPA)
- ④ Chosen Ciphertext Attack (CCA)



IND-EAV Security Game (we have already seen)

- Adversary $\mathcal{A}$ chooses two equal-length messages m_0, m_1 .
- Challenger samples $k \leftarrow \text{Gen}(1^n)$ and $b \leftarrow \{0, 1\}$.
- Challenger returns $c \leftarrow \text{Enc}_k(m_b)$.
- Adversary outputs a guess b' .

$$\text{Adv}_{\Pi}^{\text{eav}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$



IND-CPA Security Game

- Adversary has access to an encryption oracle $\text{Enc}_k(\cdot)$.
- Adversary chooses m_0, m_1 .
- Challenger encrypts one of them.
- Adversary continues using the encryption oracle.
- Adversary outputs b' .

$$\text{Adv}_{\Pi}^{\text{cpa}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$



IND-CCA1 Security Game

- Adversary has encryption and decryption oracles.
- Decryption oracle is available only before the challenge.
- Challenger encrypts one challenge message.
- Adversary outputs b' .



IND-CCA2 Security Game

- Adversary has encryption oracle.
- Adversary has decryption oracle before and after challenge.
- Only restriction:

c^*

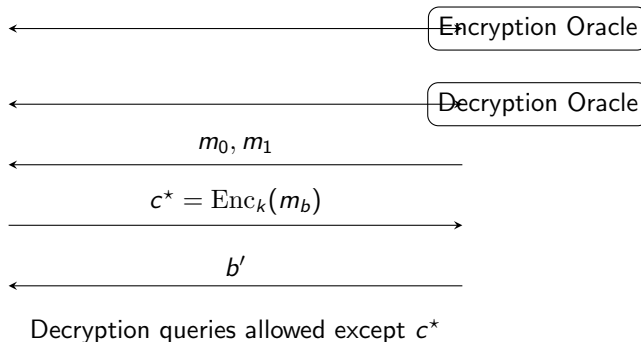
(the challenge ciphertext) cannot be queried.



IND-CCA2 Security Game (contd)

Challenger

Adversary



$$\text{Adv}_{\Pi}^{\text{cca}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$



Hierarchy of Security Notions

$\text{IND-CCA2} \implies \text{IND-CCA1} \implies \text{IND-CPA} \implies \text{IND-EAV}$

Notion	Attacker Capability
IND-EAV	Challenge ciphertext only
IND-CPA	Encryption oracle
IND-CCA1	Decryption oracle before challenge
IND-CCA2	Decryption oracle before and after challenge



Key Takeaways

- Perfect secrecy is achievable via OTP.
- Perfect secrecy is expensive.
- Modern cryptography uses computational assumptions.
- Security is defined through experiments.
- IND-CPA is the minimum standard for encryption.
- IND-CCA provides stronger guarantees.



What Symmetric Cryptography Cannot Do

Symmetric cryptography alone cannot provide:

- Public verification
- Digital signatures
- Scalable key distribution
- Non-repudiation

Motivation for public-key cryptography.



The Public-Key Revolution

Introduced by:

- Diffie and Hellman (1976) (Key exchange, useful in Symmetric Key Crypto, as we need shared keys)
- Rivest, Shamir and Adleman (1977) (Public Key Cryptography)

Fundamental idea of Public Key Cryptography:

$$pk \neq sk.$$

Encryption key can be public, but the decryption key is private.



Diffie–Hellman Key Exchange

Goal: Allow two parties to establish a shared secret over an insecure channel.

Public Parameters

p = large prime, g = generator

Alice

- Chooses private key a
- Computes

$$A = g^a \bmod p$$

- Sends A to Bob

Bob

- Chooses private key b
- Computes

$$B = g^b \bmod p$$

- Sends B to Alice

Shared Secret

$$K = B^a \bmod p = A^b \bmod p = g^{ab} \bmod p$$

Security Basis: Computing g^{ab} from public values (g, p, A, B) is computationally hard due to the *Discrete Logarithm Problem*.



Thank You

Questions?

